CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

Program 12: WRITE THE PYTHON POGRAM FOOR TIC TAC TOE GAME

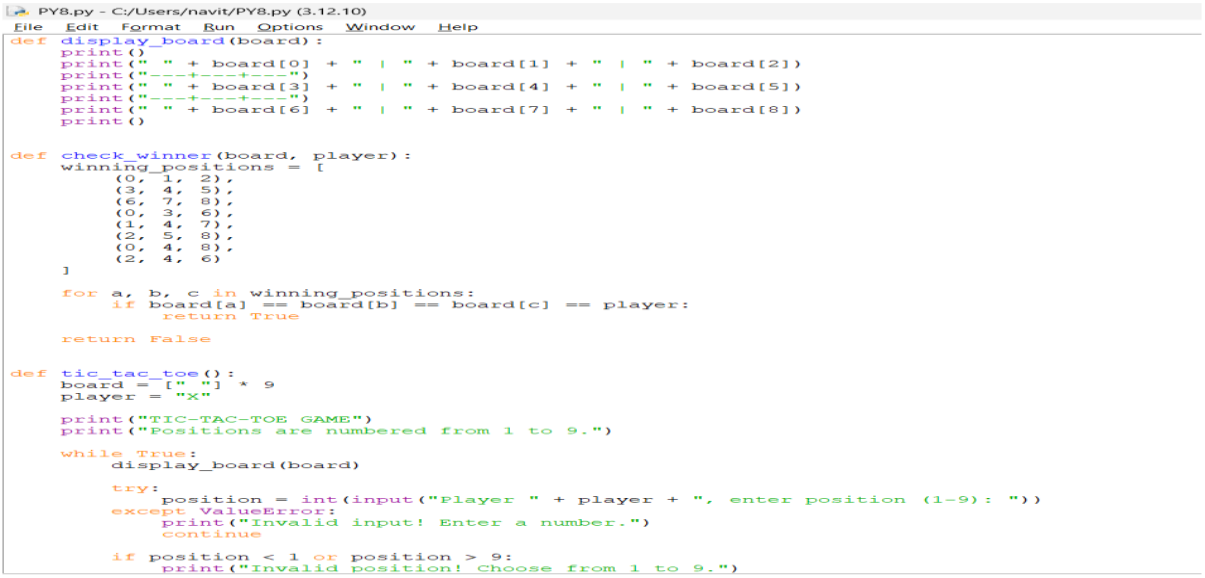
AIM

To write a Python program to implement the Tic-Tac-Toe game for two players.

ALGORITHM

1. Start the program.
2. Create a 3 × 3 game board initialized with empty spaces.
3. Display the game board.
4. Ask the current player to enter a position between 1 and 9.
5. Check whether the selected position is valid and empty.
6. Place the current player's symbol (X or O) in the selected position.
7. Check whether the current player has won by completing a row, column, or diagonal.
8. If a player wins, display the winner and stop the game.

9. If the board is full and nobody wins, declare a draw.
10. Otherwise, switch players and repeat the process.
11. Stop the program.



```
PY8.py - C:/Users/navit/PY8.py (3.12.10)
File  Edit  Format  Run  Options  Window  Help
def display_board(board):
    print()
    print(" " + board[0] + " | " + board[1] + " | " + board[2])
    print("---+---+---")
    print(" " + board[3] + " | " + board[4] + " | " + board[5])
    print("---+---+---")
    print(" " + board[6] + " | " + board[7] + " | " + board[8])
    print()

def check_winner(board, player):
    winning_positions = [
        (0, 1, 2),
        (3, 4, 5),
        (6, 7, 8),
        (0, 3, 6),
        (1, 4, 7),
        (2, 5, 8),
        (0, 4, 8),
        (2, 4, 6)
    ]
    for a, b, c in winning_positions:
        if board[a] == board[b] == board[c] == player:
            return True
    return False

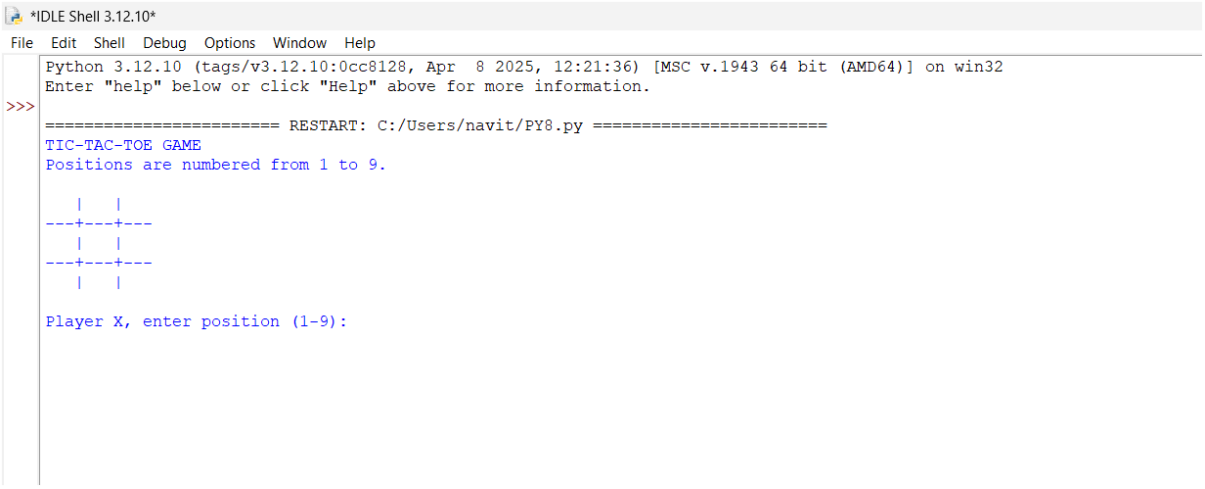
def tic_tac_toe():
    board = [" "] * 9
    player = "X"

    print("TIC-TAC-TOE GAME")
    print("Positions are numbered from 1 to 9.")

    while True:
        display_board(board)

        try:
            position = int(input("Player " + player + ", enter position (1-9): "))
        except ValueError:
            print("Invalid input! Enter a number.")
            continue

        if position < 1 or position > 9:
            print("Invalid position! Choose from 1 to 9.")
```



```
*IDLE Shell 3.12.10*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/navit/PY8.py ==========
TIC-TAC-TOE GAME
Positions are numbered from 1 to 9.

   |   |
---+---+---
   |   |
---+---+---
   |   |

Player X, enter position (1-9):
```

RESULT

The Python program was executed successfully, implementing the Tic-Tac-Toe game for two players. The game checks for a win, a draw, and invalid moves.